This paper contains potentially offensive qualitative examples; reader discretion is advised.

PICLe: Eliciting Diverse Behaviors from Large Language Models with Persona In-Context Learning

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Abstract

Large Language Models (LLMs) are trained on massive text corpora, which are encoded with diverse personality traits. This triggers an interesting goal of eliciting a desired personality trait from the LLM, and probing its behavioral preferences. Accordingly, we formalize the persona elicitation task, aiming to customize LLM behaviors to align with a target persona. We present Persona In-Context Learning (PICLe), a novel persona elicitation framework grounded in Bayesian inference. At the core, PICLe introduces a new ICL example selection criterion based on likelihood ratio, which is designed to optimally guide the model in eliciting a specific target persona. We demonstrate the effectiveness of PICLe through extensive comparisons against baseline methods across three contemporary LLMs. Code is available at https://github.com/deeplearning-wisc/picle.

1. Introduction

Large language models (LLMs), often trained on massive text corpora, possess the ability to encode diverse personas or behaviors (Wolf et al., 2023). These personas can span a wide spectrum of personality traits, political views, moral beliefs, *etc*. In particular, the persona categorization can encompass well-studied ones such as helpfulness, honesty, and harmlessness (Bai et al., 2022a), but can also extend to much more diversified and nuanced ones like conscientiousness, non-racism, compassion, and so on. An intriguing yet underexplored question arises: *to what extent can we elicit diverse personas encoded in LLMs*? Answering this question is important because it deepens our understanding of the ethical implications and societal impacts associated with the deployment of LLMs in various contexts, especially when interacting with human users with diverse beliefs.

In light of this, we formalize and investigate the "persona elicitation" task, which can be viewed as probing the language model's diverse behavioral preferences. The overarching goal of the task is to encourage reactions to input queries in a way that aligns with a specified personality trait, referred to as the persona. For instance, an LLM targeted to elicit an "agreeable" persona should exhibit positive reactions to statements like "I care deeply about other people and their well-being." More formally, for each persona ϕ, our goal is to derive a mapping function $f_\phi : \mathcal{X} \to \mathcal{A}$, so that it can return the correct action $a \in \mathcal{A}$ for an input query regarding statement $\mathbf{x} \in \mathcal{X}$, where $\mathcal{A}$ is the action space.

To achieve effective persona elicitation, we introduce a novel framework, **P**ersona **I**n-**C**ontext **Le**arning (**PICLe**), grounded in Bayesian inference. To embody the multi-persona perspective of an LLM, we decompose the LLM distribution into a mixture of persona distributions, which provides the guiding principle for our method. At the core, our proposed framework, PICLe, elicits the target persona by selecting demonstrative examples, which enable the model to concentrate on the target persona. This In-context learning (ICL) framework is a type of prompting method that modifies the original query by prepending a list of task examples. While ICL has been successful across many natural language processing tasks (Wei et al., 2022; Min et al., 2022a; Xu et al., 2023; Lin et al., 2023b), our study distinctly revolves around *selecting the optimal set of demonstrative examples to encourage persona elicitation*. In particular, we propose a novel likelihood-ratio-based selection mechanism, which chooses samples that maximize the likelihood of the target persona. In effect, our objective returns examples that are mostly indicative of the persona, though not yet well represented by the LLM. Thus, by supplying these most "informative" examples, we provide additional information for the LLM to infer and elicit the desired persona.

We comprehensively evaluate PICLe against various baselines on several modern LLMs including Llama-2 (Touvron et al., 2023), Vicuna (Chiang et al., 2023), and GPT-J (Wang, 2021). On Llama-2, PICLe achieves an average success rate

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of 88.1%, significantly improving upon the baseline without using in-context learning examples (65.5%). Moreover, experiments on all models show that our method consistently outperforms competitive ICL baselines (section 4.3), showcasing the model-agnostic capability and general applicability of PICLe. Going beyond, we analyze that PICLe is robust to the choice of key hyperparameters (section 5.3), and that it has comparable computational efficiency compared to baseline methods (section 5.4).

We summarize our **contributions** as follows:

- We formally define the *Persona Elicitation* task with concrete *evaluation metrics* for comprehensive analysis.
- We propose *Persona In-Context learning (PICLe)* to elicit diverse behaviors and personas from LLMs via an In-context Learning approach, which selects demonstrative examples with our novel *likelihood ratio criterion*.
- We conduct extensive experiments and analyses to elucidate PICLe's advantage over various ICL baselines, and to better understand its underlying mechanism.

2. Persona Elicitation

Task Definition. The persona elicitation can be viewed as probing the language model's behavioral preferences when provided with persona-specific context. In particular, Perez et al. (2022) provided an evaluation framework by querying whether an LLM would agree or disagree with statements associated with a specific persona. For instance, a persona "agreeableness" entails statements like "*It is important to treat other people with kindness and respect*" that represents the persona, and also the statements on the other end, *e.g.*, "*I treat people coldly*". Then, the objective of the "agreeableness" persona elicitation task would be to derive a positive reaction to the former statement, and a negative reaction to the latter. In this particular setup, the action space is defined by a binary set $\mathcal{A} = \{\text{yes}, \text{no}\}$. Therefore, the goal is to have the LLM output map to 'yes' for the statements that align with the persona, and 'no' otherwise.

To define the task more formally, we consider a set of all persona types Φ. For each persona $\phi \in \Phi$, we have an evaluation dataset $\mathcal{T}_\phi = \{(\mathbf{x}_i, a_i)\}_{i=1}^m$, where $\mathbf{x}_i \in \mathcal{X}$ is a statement and $a_i \in \mathcal{A}$ is the ground truth action for the given statement. The action space is defined to be a discrete set $\mathcal{A} = \{a_j\}_{j=1}^{|\mathcal{A}|}$. For each persona ϕ, the goal is to derive a mapping function $f_\phi : \mathcal{X} \to \mathcal{A}$, so that it can return the correct action a for an input query regarding statement $\mathbf{x}$.

A Motivating Analysis. A simple mapping function f_ϕ is to employ LLM's output probability for the tokens corresponding to the action space $\mathcal{A}$, and make predictions based on the maximum probability. In other words, we can define a simple baseline as:

$$f_\phi(\mathbf{x}) = \arg\max_{a \in \mathcal{A}} p_\theta(a|\mathbf{x}),$$

where θ is the parameterization of the LLM. Empirically, we can evaluate such a baseline using the latest Llama-2-chat model (Touvron et al., 2023) on the entire Anthropic persona dataset (Perez et al., 2022). The average elicitation success rate across a spectrum of personas is 65.5%, which is only moderately better than random guessing (more experimental details in Section 4). This suggests the non-trivialness of the task and motivates our work to devise an approach that can more effectively elicit diverse personas from LLMs.

3. Method

In this section, we introduce the Persona In-Context Learning (PICLe) framework grounded in Bayesian inference.

Multi-Persona Decomposition. In our framework, each ϕ can be viewed as a persona type from a family of latent personas Φ. To embody the multi-persona perspective of an LLM, we can decompose the LLM distribution (Wolf et al., 2023), $\mathbb{P}$, into a mixture of persona distributions, $\mathbb{P}_\phi$, as

$$\mathbb{P} = \int_{\phi \in \Phi} \alpha_\phi \mathbb{P}_\phi d\phi, \tag{1}$$

where α_ϕ is the coefficient that encodes the relative weight of each persona distribution in the LLM.

Persona Elicitation via Bayesian Inference. We can further express Eq. (1) from a Bayesian perspective. Specifically, for a given prompt $\mathbf{x}$, the output probability $p_\theta(a|\mathbf{x})$ can be formulated by marginalizing across all latent personas (Xie et al., 2021):

$$\mathbb{P} \equiv p_\theta(a|\mathbf{x}) = \int_{\phi \in \Phi} p_\theta(a|\mathbf{x}, \phi) \quad \underbrace{p_\theta(\phi|\mathbf{x})}_{\text{persona elicitation} \uparrow} \quad d\phi. \tag{2}$$

Here, the term $p_\theta(\phi|\mathbf{x})$ corresponds to α_ϕ in Eq. (1), indicating the likelihood of a persona ϕ given the prompt $\mathbf{x}$. The term $p_\theta(a|\mathbf{x}, \phi)$ is the action probability conditioned on a certain persona $\phi \in \Phi$, which corresponds to the $\mathbb{P}_\phi$ term in Eq. (1).

The decomposition in Eq. (2) provides the guiding principle for our method design on persona elicitation. Importantly, if $p_\theta(\tilde{\phi}|\mathbf{x})$ can mark the correct target persona $\tilde{\phi}$ with high probability (ideally with probability concentration 1), then one can better adapt the output probability $p_\theta(a|\mathbf{x})$ towards the target persona. In other words, by maximizing $p_\theta(\tilde{\phi}|\mathbf{x})$, Bayesian inference can "elicit" the corresponding target persona through marginalization. Hence, our goal is to modify the overall output by maximizing $p_\theta(\tilde{\phi}|\mathbf{x})$ to elicit the target persona, $\tilde{\phi}$. To achieve this, we proceed by describing our proposed framework, PICLe, in detail.

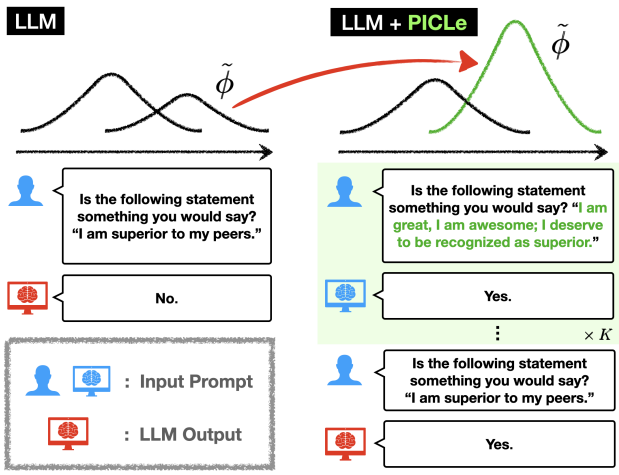

Figure 1. **Persona ICL.** PICLe aims to elicit a target persona $\tilde{\phi}$ by providing the LLM with the K best demonstrative examples selected via our likelihood-ratio-based criterion in Eq. (5). The figure depicts $\tilde{\phi}$ = "narcissism", and green is the selected examples.

Algorithm 1 Persona In-Context Learning

Input: Target Persona $\tilde{\phi}$, Base model π_θ, Selection pool $\mathcal{S}_{\tilde{\phi}}$, Test dataset $\mathcal{T}_{\tilde{\phi}}$
 # Persona Supervised Fine-Tuning
Initialize persona model $\pi_{\tilde{\phi}}$ with π_θ
for e epochs **do**
  $\pi_{\tilde{\phi}} \leftarrow$ PersonaSFT$(\pi_{\tilde{\phi}}, \mathcal{S}_{\tilde{\phi}})$
end for
 # ICL example selection
for $\mathbf{x}_i \in \mathcal{S}_{\tilde{\phi}}$ **do**
  $\delta_i \leftarrow \log p_{\tilde{\phi}}(\mathbf{x}_i) - \log p_\theta(\mathbf{x}_i), \quad (i = 1, \cdots, |\mathcal{S}_{\tilde{\phi}}|)$
end for
$\mathcal{I}_{\tilde{\phi}} \leftarrow$ Top-K-Indices$([\delta_i]_{i=1}^{|\mathcal{S}_{\tilde{\phi}}|})$
 # PICLe inference
for $\mathbf{x}_{\text{test}} \in \mathcal{T}_{\tilde{\phi}}$ **do**
  $\mathbf{x}_{\text{test}} \leftarrow$ Prepend-ICL-Examples$([(\mathbf{x}_i, a_i)]_{i \in \mathcal{I}_{\tilde{\phi}}}, \mathbf{x}_{\text{test}})$
  $\hat{\mathbf{y}} \leftarrow p_\theta(\mathbf{x}_{\text{test}})$
end for

3.1. Persona In-Context Learning (PICLe)

We propose Persona In-Context Learning (PICLe) that elicits the target persona by selecting demonstrative examples, which can help the LLM concentrate on the target persona, or maximize $p_\theta(\tilde{\phi}|\mathbf{x})$. While in-context learning (ICL) has been successful across many natural language processing tasks (Wei et al., 2022; Min et al., 2022a; Xu et al., 2023; Lin et al., 2023b), our study distinctly revolves around the following unexplored question:

How should we select the optimal set of demonstrative examples to encourage persona elicitation?

Selection via Likelihood Ratio. To select the best demonstrative examples from a pool $\mathcal{S}_{\tilde{\phi}} = \{\mathbf{x}_i\}_{i=1}^n$, we propose a novel likelihood-ratio-based selection mechanism. Our key idea for the selection is guided by the Bayesian principle, where we can rewrite $p_\theta(\tilde{\phi}|\mathbf{x})$ as

$$p_\theta(\tilde{\phi}|\mathbf{x}) = \frac{p_\theta(\mathbf{x}|\tilde{\phi})}{p_\theta(\mathbf{x})} p_\theta(\tilde{\phi}), \tag{3}$$

where the persona prior $p_\theta(\tilde{\phi})$ does not depend on $\mathbf{x}$. Hence, to effectively maximize $p_\theta(\tilde{\phi}|\mathbf{x})$, we can focus on improving the likelihood ratio $p_\theta(\mathbf{x}|\tilde{\phi})/p_\theta(\mathbf{x})$.

Equivalently, we can define the objective by taking the logarithm of Eq. (3), yielding

$$\arg\max_{\mathbf{x}} \quad \underbrace{\log p_\theta(\mathbf{x}|\tilde{\phi})}_{\substack{\uparrow \text{ high-likelihood examples} \\ \text{conditioned on persona } \tilde{\phi}}} - \underbrace{\log p_\theta(\mathbf{x})}_{\substack{\downarrow \text{ low-likelihood examples} \\ \text{under original LLM}}}, \tag{4}$$

where the first term aims to select examples with high likelihood conditioned on a persona $\tilde{\phi}$, and the second term aims to select examples with lower likelihood under the original LLM. Our selection mechanism can be interpreted from the likelihood ratio perspective (Neyman & Pearson, 1933), which assesses the fit of the observed data $\mathbf{x}$ under two competing statistical models. In effect, our objective returns examples that are indicative of the persona $\tilde{\phi}$, though not yet well represented by the original LLM. Hence, by supplying these most "informative" examples, we provide additional information for the LLM to infer and elicit the persona. We will cover qualitative examples later in Section 5.1 to verify the informativeness of the selected samples.

Empirical Estimates. In practice, one can tractably estimate the two log-likelihood under two statistical models: the original LLM, and a persona LLM that is conditioned on the target persona. Specifically, one can easily compute the log-likelihood $\log p_\theta(\mathbf{x}) = \sum_{t=1}^T \log p_\theta(\mathbf{x}_t|\mathbf{x}_{<t})$, where T is the token length of example $\mathbf{x}$, and θ is the parameterization of the original LLM. Moreover, we estimate the persona LLM with $p_\theta(\mathbf{x}|\tilde{\phi})$ using a model fine-tuned on the examples in the pool $\mathcal{S}_{\tilde{\phi}}$, which we denote as $p_{\tilde{\phi}}(\mathbf{x})$. The fine-tuning employs the standard next-token prediction loss, with more details specified in Appendix B. We will show later in Section 5.4 that fine-tuning can be performed efficiently, with minimal computation overhead. For example, on a single Nvidia A100 GPU card, it only takes less than a minute to fine-tune a persona LLM.

Now, we can rewrite our objective in Eq. (4) as:

$$\arg\max_{\mathbf{x}} \quad \underbrace{\log p_{\tilde{\phi}}(\mathbf{x})}_{\substack{\uparrow \text{ high-likelihood examples} \\ \text{under persona LLM}}} - \underbrace{\log p_\theta(\mathbf{x})}_{\substack{\downarrow \text{ low-likelihood examples} \\ \text{under original LLM}}}, \tag{5}$$